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Bihar Engineering University, Patna

B.Tech 2nd Semester Exam., 2022

Code: 105201 | ***Subject: Physics (Semiconductor Physics and Introduction to Quantum Mechanics)*** | ***Time: 3 hours*** | ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*
(v) *Symbols used (if any) have their usual meanings.*

Q.1 Answer any seven questions : **[2×7=14]**

(a) What is the unit of mobility of charge carriers?

(b) Give examples of direct band semiconductor.

(c) What is de Broglie wavelength of an electron which has been accelerated from rest through a potential difference of 100 V? Given: Mass of electron = $9{\cdot}1\times10^{-31}$ kg, Planck's constant = $6{\cdot}62\times10^{-34}$ Js

(d) Define Fermi level.

(e) Define expectation value.

(f) What are the types of photodetectors?

(g) Find uncertainty in the momentum of a particle when its position is determined within 0·01 cm.

(h) Define population inversion.

(i) What do you mean by diffusion in carrier transport?

(j) Write two advantages of LED over ordinary incandescent lamp.

Q.2 Solve both questions : **[10+4=14]**

(a) Derive the expression for equilibrium concentration of electrons in conduction band.

(b) Calculate the equilibrium concentration of electrons in silicon at T = 300K. Assume, the Fermi energy is 0.25 eV below the conduction band. The value of effective density of state function in conduction (N_c) for silicon at 300 K = $2{\cdot}8\times10^{19}cm^{-3}$, value of kT at 300 K = 0·0259eV.

Q.3 Solve this question : **[14]**

Discuss the Kronig-Penney model for the motion of an electron in a periodic potential. Plot total energy of an electron in a periodic potential versus wave number, i.e., $\varepsilon - k$ graph.

Q.4 Write short notes on the following: **[7+7=14]**

(a) Compton effect

(b) Photoelectric effect

Q.5 Solve this question : **[14]**

Discuss the structure, working principle and characteristics of PIN and Avalanche photodiodes.

Q.6 Solve this question : **[14]**

What is non-radiative recombination mechanism? Discuss the processes involved in non-radiative recombination in detail.

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Q.7 Solve this questions : **[10+4=14]**

(a) Derive an expression for effective mass of an electron.
(b) What is the significance of negative effective mass?

Q.8 Solve this question : **[14]**

What are LEDs? Sketch the relative eye response as a function of wavelength. Describe device structure and materials used for LED with its characteristics.

Q.9 Write short notes on the following: **[7+7=14]**

(a) Density of states

(b) Free electron model

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